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# MCS–011

## MASTER OF COMPUTER APPLICATIONS/BACHELOR OF COMPUTER APPLICATIONS (REVISED)

## (MCA/BCA)

## Term-End Examination

## December, 2025

### MCS-011 : PROBLEM SOLVING AND PROGRAMMING

*Time : 3 Hours* *Maximum Marks : 100*

*Weightage : 70%*

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***Note :*** *(i) Question No.* ***1*** *is compulsory.*

*(ii) Answer any* ***three*** *questions from the rest.*

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1. (a) Write an algorithm and draw a corresponding flowchart to find the reverse of a given number and check if it is a palindrome or not. 10

   (b) List and explain various categories of operators in a C along with an example for each. 10

   (c) Write a C program to generate the following pattern : 10

```
*
* *
* * *
* * * *
* * * * *
```

   (d) Write a C program to find the factorial of a given number using a recursive function. 10



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2. (a) What are the differences between a while loop and a do---while loop in C ? Provide C code to illustrate your explanation for both. 6

   (b) Write a C program to search a given element in the array. If value exists, then print that value along with its position (index) in the array otherwise display "element not found". 8

   (c) Differentiate between call by value and call by reference with an example. 6

3. (a) Write a program in C to classify students' performance based on their marks into different categories such as Excellent, Good, Satisfactory, Pass and Fail. 7

   *Criteria* :

   - Excellent : Marks greater than or equal to 90
   - Good : Marks between 80 and 89



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- Satisfactory : Marks between 70 and 79
- Pass : Marks between 40 and 69
- Fail : Marks less than 40

(b) Explain the concept of preprocessor directives in C programming. Provide examples of *three* commonly used preprocessor directives and describe their functions. 8

(c) Describe file handling in C. Explain various access modes for opening the file. 5

4. (a) What will be the output of the following code ? Also, explain the reason for that output : 5

```
#include<stdio.h>
int main ( ) {
    int x = 5, y = 3, z ;
    z = x – – * y;
    printf ("%d, %d", x, z);
    return 0;
}
```



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(b) Write a C program to check if a given number is Armstrong number or not.

(An Armstrong number is a special kind of number in Math. It's a number that equals the sum of its digits, each raised to a power $n$, where $n$ is the number of digits in the number for example, if you have a number like 153, it's an Armstrong number because $1^3 + 5^3 + 3^3$ equals 153). 10

(c) Define an array. How are arrays declared and initialized ? Write a C code segment to explain it. 5

5. (a) Differentiate between *break* statement, *continue* statement and *goto* statement, with examples. 10



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(b) Explain any *five* built-in string functions in C with examples. 5

(c) Differentiate between Structure and Union. Write a C code segment to explain the difference. 5

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